Vitali–Carathéodory does not force Newtonian quasicontinuity

Exact affirmative answer in later literature

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Status. `literature_already_answered`. This packet identifies and checks an exact later-literature answer; it does not claim a new theorem.

Source question

Björn, Björn, and Malý, *Non-quasicontinuous Newtonian functions and outer capacities based on Banach function spaces*, arXiv:2503.21665, study the following properties of a Newtonian space $N^1X(\mathcal{P})$:

$$\begin{aligned}
(A):&\ \text{every Newtonian function is quasicontinuous,}\\
(B):&\ C_X \text{ is an outer capacity,}\\
(D):&\ \text{weak quasicontinuity implies quasicontinuity,}\\
(F):&\ \text{every Newtonian class has a quasicontinuous representative,}\\
(G):&\ \text{every Newtonian function is weakly quasicontinuous.}
\end{aligned}$$

Their Theorem 1.5 proves (B) for proper $\mathcal{P}$ and (D) for locally compact $\mathcal{P}$ when X has the Vitali–Carathéodory property. Immediately afterward (PDF page 4), they ask whether the remaining properties can fail under that property. In the locally compact situation, the genuinely remaining quasicontinuity assertions are $(A), (F), (G)$.

Exact later answer

Björn and Björn, *Quasicontinuity of $N^{1,\infty}$ functions and the Vitali–Carathéodory property on general metric spaces*, arXiv:2605.22674, answer this existential question affirmatively. Their Theorem 1.2 states that, under Vitali–Carathéodory for $L^\infty(\mathcal{P})$, C_∞ is outer and

$$(A) \iff (F) \iff (G),$$

but examples exist in which all three assertions fail.

Their Example 3.3 makes the answer completely explicit:

$$\mathcal{P} = \{0, 2^{-n} : n \geq 1\}, \qquad \mu = \delta_0 + \sum_{n=1}^{\infty} 2^{-n} \delta_{2^{-n}}, \qquad X = L^\infty(\mathcal{P}).$$

The space is compact, hence proper and locally compact. Nevertheless, $\chi_{\{0\}} \in N^{1,\infty}(\mathcal{P})$ is not weakly quasicontinuous and has no quasicontinuous representative. Thus $(A), (F), (G)$ fail even though $L^\infty(\mathcal{P})$ has the Vitali–Carathéodory property.

Why the answer is exact

The source asks an existence question over function spaces having the Vitali–Carathéodory property. The later example uses L^∞, which is a Banach function space in this atomic setting, and the metric space is even compact. It therefore supplies precisely the counterexample requested for the remaining properties, rather than merely weakening one of the source hypotheses.

This answer has limited scope. It does not settle the separate implication $(F) \Rightarrow (E)$ for Banach function spaces, nor the general removal of properness in Theorem 1.5. The later paper does remove properness for the special case $X = L^\infty$.

Proof Intuition

The construction is a convergent sequence with positive mass at every point. Positive point masses make essential suprema genuine pointwise suprema; this yields the Vitali–Carathéodory property. The space has no nonconstant curves, and every nonempty set has C_∞-capacity one, so the capacity is outer and there are no nonempty capacity-zero exceptional sets.

But 0 is a nonisolated point. Hence $\chi_{\{0\}}$ is discontinuous there. Because an exceptional set of capacity less than one must be empty, weak quasicontinuity is just continuity. Likewise, equality quasi-everywhere is literal equality everywhere. Consequently no representative can repair $\chi_{\{0\}}$, forcing the simultaneous failure of $(A), (F), (G)$.

References and provenance

[1] A. Björn, J. Björn, and L. Malý, *Non-quasicontinuous Newtonian functions and outer capacities based on Banach function spaces*, arXiv:2503.21665; source question on PDF page 4.

[2] A. Björn and J. Björn, *Quasicontinuity of $N^{1,\infty}$ functions and the Vitali–Carathéodory property on general metric spaces*, arXiv:2605.22674; Theorem 1.2 on PDF page 2 and Example 3.3 on PDF page 4.

The bundled PDFs are official arXiv renderings. Their SHA-256 values are `3e35c858...ab480` and `a4941f1e...44f5`, respectively.